

AquaPulse — 2-Minute Official Pitch & Code Walkthrough

Prepared by Team Mythos for Ministry of Jal Shakti / CGWB Evaluation Panel

Team: Team Mythos	SIH Category: Software
Problem ID: SIH25068	Ministry: Ministry of Jal Shakti (CGWB)
Problem Statement: Real-time Groundwater resource evaluation using DWLR data	Status: Verified & Deployed
Live Demo: https://savanrooj.github.io/mythos/	GitHub: https://github.com/savanrooj/mythos

1. The 30-Second Elevator Pitch (The Hook)

'Respected judges, groundwater sustains 85% of rural India and 60% of our agriculture. While CGWB operates over 15,000 telemetric DWLR piezometers, groundwater assessments have traditionally been retrospective—meaning critical localized over-extraction is identified months too late.

*We built **AquaPulse**: an intelligent, real-time groundwater resource evaluation platform that ingests hourly telemetry streams, evaluates aquifer stress using CGWB-standard velocity and baseline metrics, predicts water levels 30 days ahead using Holt Exponential Smoothing, and instantly routes emergency SMS/Email alerts to District Magistrates before critical aquifer drawdown occurs.'*

2. Official 2-Minute Stage Demo Script (Click-Through Guide)

Time	Stage Action	Exact Spoken Lines for Judges
0:00 to 0:25	Start on National Overview . Hover over State Stress Bar & 12m Drying Trend.	<i>'Here on our National Overview, administrators see instant state-wise stress distributions, track national water table decline velocity over 12 months, and immediately identify the top 10 most at-risk stations in India.'</i>
0:25 to 0:55	Click #1 Latur Over-Extract . App opens Map + Detail Drawer. Toggle 90d, show 30d forecast band, click CSV .	<i>'Drilling into Latur reveals our 2-metric engine evaluating baseline deviation vs. 30-day decline rates. Using Holt Exponential Smoothing, AquaPulse projects water table depth 30 days into the future with 95% confidence bands.'</i>
0:55 to 1:25	Click Alerts tab in sidebar. Filter by Status Downgrade . Click Alert SMS/Mail . Show dispatch modal.	<i>'Telemetry is noisy. Our engine flags gaps, spikes, and downgrades. With one click, our automated dispatch gateway routes targeted SMS alerts to District Hydrologists and official NIC emails to District Magistrates.'</i>
1:25 to 1:45	Click Simulate Tick in header. Watch toast & live reading counter update.	<i>'Clicking Simulate Tick ingests a new hourly reading packet across all 30 stations, advances our virtual telemetry timeline, and re-evaluates stress classifications in sub-second time.'</i>
1:45 to 2:00	Click How It Works in header. Show plain-language guide modal & close.	<i>'AquaPulse is production-ready. Through our plug-and-play DataSourceAdapter, connecting to live India-WRIS APIs requires zero code changes. AquaPulse turns raw telemetry into proactive water security. Thank you!'</i>

3. The Code & Technology Explained Simply

For non-technical judges or quick technical deep-dives, here is how each module is engineered:

Component	How It Works in Plain Language	Key File in Repository
Stress Classifier	Takes the worse of Metric A (how deep compared to 12m avg) and Metric B (how fast it is falling over 30 days) to categorize as Safe, Semi-Critical, Critical, or Over-Exploited.	backend/evaluation.py
Holt Forecasting	A mathematical forecasting model that tracks both current water level ($\alpha=0.3$) and drawdown trajectory ($\beta=0.1$) to project 30 days ahead with 95% statistical confidence bounds.	backend/forecasting.py
Anomaly Detector	Scans incoming data for >6h missing gaps, >5m electrical spikes, rapid 7-day declines (>0.05m/d), and classification downgrades, generating actionable alert records.	backend/anomaly.py
India-WRIS Adapter	Uses the Strategy Design Pattern. The rest of the app doesn't care whether data comes from SQLite or the live India-WRIS REST API; only this one file is swapped.	backend/data_source.py
Frontend Web App	Built with React 18 and Vite with Leaflet map markers and Recharts visualizer. Features an in-browser fallback engine allowing 100% standalone execution on GitHub Pages.	frontend/src/App.jsx

4. Judge Q&A; Defense Strategy (Winning Answers)

Judge Question	Winning Response for Team Mythos
Why Holt smoothing instead of LSTM deep learning?	<i>'Holt smoothing gives exact mathematical interpretability, runs in sub-milliseconds on low-cost edge servers, and avoids requiring thousands of historical training epochs on newly installed DWLR sensors.'</i>
How does your system handle missing sensor packets?	<i>'Minor dropouts (<6 hours) are interpolated for rolling statistics. If a sensor drops data for >6 consecutive hours, our Gap Detector triggers a WARNING alert for field sensor maintenance.'</i>
How will you scale to 15,000+ national DWLR stations?	<i>'Our backend is decoupled via the Strategy Pattern. In production, we deploy an Apache Kafka/MQTT ingestion pipeline with TimescaleDB, handling 15,000+ telemetry packets per second.'</i>
What is the social and economic impact?	<i>'By transforming retrospective annual assessments into real-time alerts, District Magistrates can enforce agricultural extraction limits weeks earlier, safeguarding drinking water for rural communities.'</i>